1. Item 1: Perform a simple calculation

- **Highly relevant**: The question requires calculating the average step length based on the total number of steps (22,500) and the total distance (9 km), which is essentially a division operation:

$$\text{Step length} = \frac{9000\ \text{m}}{22,500} \times 100\ \text{cm/m}$$

- **Analysis**: This is the most direct mathematical execution, a basic arithmetic operation.

2. Item 12: Use a multi-step procedure leading to a mathematical solution, conclusion, or generalization

- **Highly relevant**: Solving the problem requires a multi-step process:
  1. Convert 9 km to meters (9000 m).
  2. Convert meters to centimeters (9000 × 100 = 900,000 cm).
  3. Calculate the average step length: 900,000 cm ÷ 22,500.
- **Analysis**: This process includes unit conversion and division, making it a typical example of multi-step problem-solving.

3. Item 13: Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy

- **Relevant**: The question requires the answer in centimeters, and the step length is an estimate, so high precision is not needed.
- **Analysis**: Understanding the requirements for "estimation" and "units" guides the calculation process and result presentation.

4. Item 5: Understand and utilize constructs based on definitions, rules, and formal systems, including employing familiar algorithms

- **Relevant**: The standard algorithm "average step length = total distance ÷ total number of steps" is used, along with unit conversion rules (1 m = 100 cm, 1 km = 1000 m).
- **Analysis**: The calculation relies on mathematical definitions and conversion rules.

5. Item 7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **Weakly relevant**: Involves numerical operations (division, multiplication) and unit conversion, which fall under numerical manipulation.
- **Analysis**: Although no graphs or algebra are involved, this is essentially the "manipulation" of numerical operations.

6. Item 8: Articulate a solution, showing and/or summarizing and presenting intermediate mathematical results

- **Weakly relevant**: If the problem-solving process includes showing unit conversion steps and division, it aligns with this item.
- **Analysis**: Presenting intermediate results reflects clarity in problem-solving, though the question itself does not require it.

1. 第1项：Perform a simple calculation

- **强相关**：题目要求根据总步数（22,500步）与总路程（9 km）计算平均步长，本质是除法运算：

$$步长 = \frac{9000\ \mathrm{m}}{22,500} \times 100\ \mathrm{cm/m}$$

- **分析**：这是最直接的数学执行行为，属于基础算术操作。

2. 第12项：Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- **强相关**：解题需分步执行：
  1. 将 9 km 转换为米（9000 m）
  2. 将米转换为厘米（9000 × 100 = 900,000 cm）
  3. 计算平均步长：900,000 cm ÷ 22,500
- **分析**：该过程包含单位换算与除法，是多步骤解题的典型例子。

3. 第13项：Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy

- **相关**：题目要求以厘米为单位给出答案，且步长估计为平均值，无需过高精度。
- **分析**：理解“估计”和“单位”要求，指导计算过程与结果表达。

4. 第5项：Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

- **相关**：运用“平均步长 = 总距离 ÷ 总步数”这一标准算法，并执行单位换算规则（1 m = 100 cm，1 km = 1000 m）。
- **分析**：依赖数学定义与换算规则完成计算。

5. 第7项：Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **弱相关**：涉及数字操作（除法、乘法）与单位换算，属于数值处理。
- **分析**：虽未涉及图形或代数，但本质是数值运算的“操作”。

6. 第8项：Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- **弱相关**：若在解题过程中展示单位换算步骤与除法过程，则符合此项。
- **分析**：呈现中间结果是解题清晰的体现，但题目本身不强制要求。